1 I

1.1 (1)

Using the lowering operator formula,

$$\frac{\hat{T}_-}{\hbar}|T, m_T\rangle = \sqrt{T(T+1) - m_T(m_T-1)}|T, m_T - 1\rangle,$$

we obtain

$$\begin{aligned}\frac{\hat{L}_-}{\hbar}|1,1\rangle &= \sqrt{2}|1,0\rangle,\\ \frac{\hat{L}_-}{\hbar}|1,0\rangle &= \sqrt{2}|1,-1\rangle,\\ \frac{\hat{L}_-}{\hbar}|1,-1\rangle &= 0.\end{aligned}$$

Similarly, for the spin angular momentum,

$$\begin{aligned}\frac{\hat{S}_-}{\hbar}\left|\frac{1}{2},\frac{1}{2}\right\rangle &= \left|\frac{1}{2},-\frac{1}{2}\right\rangle,\\ \frac{\hat{S}_-}{\hbar}\left|\frac{1}{2},-\frac{1}{2}\right\rangle &= 0.\end{aligned}$$

1.2 (2)

Following the hint, we start from the state with the maximum total angular momentum:

$$\left|\frac{3}{2},\frac{3}{2}\right\rangle = \left|1,1;\frac{1}{2},\frac{1}{2}\right\rangle.$$

Applying the lowering operator $\hat{J}_- = \hat{L}_- + \hat{S}_-$,

$$\frac{\hat{J}_-}{\hbar}\left|\frac{3}{2},\frac{3}{2}\right\rangle = \sqrt{3}\left|\frac{3}{2},\frac{1}{2}\right\rangle.$$

On the other hand,

$$\begin{aligned}&\frac{\hat{J}_-}{\hbar}\left|1,1;\frac{1}{2},\frac{1}{2}\right\rangle\\ &= \frac{\hat{L}_- + \hat{S}_-}{\hbar}\left|1,1;\frac{1}{2},\frac{1}{2}\right\rangle\\ &= \sqrt{2}\left|1,0;\frac{1}{2},\frac{1}{2}\right\rangle + \left|1,1;\frac{1}{2},-\frac{1}{2}\right\rangle.\end{aligned}$$

Thus,

$$\boxed{\begin{aligned}\left|\frac{3}{2},\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}&\left|1,0;\frac{1}{2},\frac{1}{2}\right\rangle\\ &+ \sqrt{\frac{1}{3}}\left|1,1;\frac{1}{2},-\frac{1}{2}\right\rangle.\end{aligned}}$$

For convenience, we will henceforth denote $|L, m_L; S, m_S\rangle$ by $|m_L, m_S\rangle$, since $L = 1$ and $S = 1/2$ remain fixed throughout this part.

Therefore, the result can be written as

$$\boxed{\left|\frac{3}{2}, \frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\left|0, \frac{1}{2}\right\rangle + \sqrt{\frac{1}{3}}\left|1, -\frac{1}{2}\right\rangle}.$$

1.3 (3)

For $J = 1/2$ and $m_J = 1/2$, the condition

$$m_J = m_L + m_S = \frac{1}{2}$$

allows only two combinations:

$$(m_L, m_S) = \left(1, -\frac{1}{2}\right), \quad \left(0, \frac{1}{2}\right).$$

Thus, we can expand the state as

$$\left|\frac{1}{2}, \frac{1}{2}\right\rangle = a\left|1, -\frac{1}{2}\right\rangle + b\left|0, \frac{1}{2}\right\rangle.$$

Since states with different values of J are orthogonal, we require

$$\left\langle\frac{3}{2}, \frac{1}{2}\middle|\frac{1}{2}, \frac{1}{2}\right\rangle = 0.$$

Using the result from (2), and choosing real coefficients a and b, we obtain

$$\sqrt{\frac{1}{3}}a + \sqrt{\frac{2}{3}}b = 0,$$

which gives

$$a = -\sqrt{2}b.$$

Meanwhile, the normalization condition is

$$a^2 + b^2 = 1.$$

Combining these conditions and choosing $a > 0$, we obtain

$$a = \frac{\sqrt{6}}{3}, \qquad b = -\frac{\sqrt{3}}{3}.$$

Therefore,

$$\boxed{\left|\frac{1}{2}, \frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\left|1, -\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}}\left|0, \frac{1}{2}\right\rangle}.$$

1.4 (4)

Following the same method as in (2), we apply the lowering operator to the result obtained in (3).

Since

$$\frac{\hat{J}_-}{\hbar}\left|\frac{1}{2},\frac{1}{2}\right\rangle = \left|\frac{1}{2},-\frac{1}{2}\right\rangle,$$

we have

$$\left|\frac{1}{2},-\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\frac{\hat{J}_-}{\hbar}\left|1,-\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}}\frac{\hat{J}_-}{\hbar}\left|0,\frac{1}{2}\right\rangle.$$

Using $\hat{J}_- = \hat{L}_- + \hat{S}_-$,

$$\frac{\hat{J}_-}{\hbar}\left|1,-\frac{1}{2}\right\rangle = \sqrt{2}\left|0,-\frac{1}{2}\right\rangle,$$
$$\frac{\hat{J}_-}{\hbar}\left|0,\frac{1}{2}\right\rangle = \sqrt{2}\left|-1,\frac{1}{2}\right\rangle + \left|0,-\frac{1}{2}\right\rangle.$$

Substituting these expressions,

$$\left|\frac{1}{2},-\frac{1}{2}\right\rangle = \left(\frac{2}{\sqrt{3}}-\frac{1}{\sqrt{3}}\right)\left|0,-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|-1,\frac{1}{2}\right\rangle.$$

Finally,

$$\boxed{\left|\frac{1}{2},-\frac{1}{2}\right\rangle = \sqrt{\frac{1}{3}}\left|0,-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|-1,\frac{1}{2}\right\rangle}.$$

2 II

2.1 (5)

For the outermost electron of a cesium atom,

$$L = 0, \qquad S = \frac{1}{2}.$$

Since the total angular momentum quantum number satisfies

$$|L-S| \le J \le L+S,$$

and

$$L+S = |L-S| = \frac{1}{2},$$

the only possible value is

$$\boxed{J = \frac{1}{2}}.$$

Therefore, the possible values of m_J are

$$\boxed{m_J = \pm\frac{1}{2}}.$$

2.2 (6)

The total atomic angular momentum is

$$\hat{\boldsymbol{F}} = \hat{\boldsymbol{I}} + \hat{\boldsymbol{J}}.$$

Since $I = 7/2$ and $J = 1/2$,

$$I + J = 4, \qquad |I - J| = 3.$$

Hence,

$$\boxed{F = 3,\ 4}.$$

2.3 (7)

The hyperfine interaction Hamiltonian is

$$\hat{H} = \frac{\kappa}{\hbar^2} \hat{\boldsymbol{I}} \cdot \hat{\boldsymbol{J}}.$$

Squaring both sides of $\hat{\boldsymbol{F}} = \hat{\boldsymbol{I}} + \hat{\boldsymbol{J}}$, and noting that the components of $\hat{\boldsymbol{I}}$ commute with those of $\hat{\boldsymbol{J}}$, we obtain

$$\hat{F}^2 = \hat{I}^2 + \hat{J}^2 + 2\hat{\boldsymbol{I}} \cdot \hat{\boldsymbol{J}}.$$

Therefore,

$$\hat{\boldsymbol{I}} \cdot \hat{\boldsymbol{J}} = \frac{1}{2} \left(\hat{F}^2 - \hat{I}^2 - \hat{J}^2 \right).$$

Substituting this into the Hamiltonian,

$$\boxed{\hat{H} = \frac{\kappa}{2\hbar^2} \left(\hat{F}^2 - \hat{I}^2 - \hat{J}^2 \right)}.$$

2.4 (8)

Since

$$\begin{aligned} \hat{F}^2 |F, m_F\rangle &= \hbar^2 F(F+1) |F, m_F\rangle, \\ \hat{I}^2 |F, m_F\rangle &= \hbar^2 I(I+1) |F, m_F\rangle, \\ \hat{J}^2 |F, m_F\rangle &= \hbar^2 J(J+1) |F, m_F\rangle, \end{aligned}$$

the corresponding energy eigenvalue is

$$E_F = \frac{\kappa}{2} \left[F(F+1) - I(I+1) - J(J+1) \right].$$

For $I = 7/2$ and $J = 1/2$,

$$I(I+1) + J(J+1) = \frac{63}{4} + \frac{3}{4} = \frac{33}{2}.$$

Thus,

$$\boxed{E_F = \frac{\kappa}{2} \left[F(F+1) - \frac{33}{2} \right]}.$$

For $F = 4$,

$$\begin{aligned} E_4 &= \frac{\kappa}{2}\left(4 \times 5 - \frac{33}{2}\right) \\ &= \frac{7\kappa}{4}. \end{aligned}$$

For $F = 3$,

$$\begin{aligned} E_3 &= \frac{\kappa}{2}\left(3 \times 4 - \frac{33}{2}\right) \\ &= -\frac{9\kappa}{4}. \end{aligned}$$

Therefore, the energy difference is

$$\begin{aligned} \Delta E &= E_4 - E_3 \\ &= \frac{7\kappa}{4} + \frac{9\kappa}{4}. \end{aligned}$$

Hence,

$$\boxed{\Delta E = 4\kappa}.$$

2.5 (9)

The frequency of the electromagnetic radiation corresponding to the hyperfine transition is

$$\nu = 9\,192\,631\,770 \text{ Hz}.$$

Using the Planck relation,

$$\Delta E = h\nu = 2\pi\hbar\nu,$$

and the result from (8),

$$4\kappa = h\nu.$$

Therefore,

$$\boxed{\kappa = \frac{h\nu}{4}}.$$

Substituting the given values,

$$\begin{aligned} \kappa &= \frac{(6.62607015 \times 10^{-34})(9\,192\,631\,770)}{4} \text{ J} \\ &\approx 1.523 \times 10^{-24} \text{ J}. \end{aligned}$$

Converting to electronvolts,

$$\begin{aligned} \kappa &= \frac{1.523 \times 10^{-24}}{1.602176634 \times 10^{-19}} \text{ eV} \\ &\approx 9.50 \times 10^{-6} \text{ eV}. \end{aligned}$$

To one significant figure,

$$\boxed{\kappa \approx 1 \times 10^{-5} \text{ eV}}.$$